

Text Processing and Regular Expression

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Word

- Words are the building blocks of language.
- Each language has a fixed number of *valid* words, aka. vocabulary.

English	520,000		English Wiktionary	Contains 520,000 gloss entries and 928,989 total entries. ^[9]
English	470,000		Webster's Third New International Dictionary and Addenda Section	Contains 470,000 entries ^[15]
English	350,000		The American Heritage Dictionary of the English Language, Third Edition	In the introduction to the 4th and 5th editions, it is mentioned that more than 10,000 words have been added, thus the total for the 5th edition will be more than 370,000 words. ^{[19][failed verification]}
English	207,016		WordNet, 3.1	As of November 2012 WordNet's latest Online-version is 3.1. The database contains 155,327 words organized in 175,979 synsets for a total of 207,016 word-sense pairs. ^[36]
English	171,476		Oxford English Dictionary, Second Edition	Oxford Dictionary has 273,000 headwords; 171,476 of them being in current use, 47,156 being obsolete words and around 9,500 derivative words included as subentries. The dictionary contains 157,000 combinations and derivatives in bold type, and 169,000 phrases and combinations in bold italic type, making a total of over 600,000 word-forms.^{[41][42]} There is one count that puts the English vocabulary at about 1 million words — but that count presumably includes words such as Latin species names, prefixed and suffixed words, scientific terminology, jargon, foreign words of extremely limited English use and technical acronyms.^{[43][44][45]}
Hindi	183,175		Wiktionary, Hindi language version	एक मुक्त शब्दकोश, जो सभी को सम्पादन का अधिकार देता है। जिसमें अनेक भाषाओं के १,८३,१७५ शब्द उपलब्ध हैं। ^[40]

Word

- How do we identify a valid word, e.g., bank?



Dictionary lookup

- What about?

- Bank
- BANK
- Banks

Can be handled
through case-folding

In general, no
dictionary manages
plural form, explicitly.

Pattern Matching

- Searching/Matching a pattern (string) is very frequent in text processing.
 - Web-search or IR based system
 - Word-processing applications.

- A simpler yet powerful solution is to use *regular expression* for the pattern matching.

Regular Expression

RE		Matches	Examples: Any text containing these symbols
<code>/bank/</code>	→	Ordered seq of chars <i>b, a, n, & k</i>	bank
<code>/Banks/</code>	→	Ordered seq of chars <i>B, a, n, k, & s</i>	Banks
<code>/[Bb]ank/</code>	→	Either uppercase or lowercase <i>b</i>	bank, Bank
<code>/[Bb]anks?/</code>	→	Char <i>s</i> is optional (0 or 1)	bank, Bank, banks, Banks
<code>/colou?r/</code>	→	Char <i>u</i> is optional (0 or 1)	color, colour

Regular Expression

RE	Matches	Examples: Any text containing these symbols
<code>/ ./</code>	→ Anything of length 1	a, A, b, c, d, D, 1, 2, :, <,
<code>/ [0123456789] /</code>	→ Any number	5, 4, 1, ...
<code>/ [abcdefg] /</code>	→ Any lowercase between <i>a</i> & <i>g</i>	a, b, c, d, e, f, g
<code>/ [a-g] /</code>	→ Any lowercase between <i>a</i> & <i>g</i>	a, b, c, d, e, f, g
<code>/ [^a] /</code>	→ Anything except lowercase <i>a</i>	A, b, c, d, D, 1, 2, :, <,
<code>/ [a^b] /</code>	→ Either <i>a</i> , <i>b</i> , or <i>^</i>	a, ^, b
<code>/ beg . n /</code>	→ Any char in between <i>beg</i> & <i>n</i>	Begin, begun, began, beg1n

Regular Expression: Kleene closure

RE		Matches	Examples: Any text containing these symbols
<code>/a[*]/</code>	→	0 or more occurrence of as	ϵ , a, aa, aaa, aaa, ...
<code>/a⁺/</code>	→	1 or more occurrence of as	a, aa, aaa, aaa, ...
<code>/[ab][*]/</code>	→	0 or more as or bs	ϵ , a, b, aaaa, abba, baab
<code>/baa⁺!/[*]</code>	→	ba followed by 1 or more as & end with !	baa!, baaaaa!,

Regular Expression: Anchors

RE	Matches	Examples: Any text containing these symbols
<code>/^The[a-zA-Z]* /</code>	→ Start of the line should be <i>The</i>	The dog ...
<code>/\.\$ /</code>	→ Line must ends with a period (.)	The dog barks. , He went.
<code>/\bun\b /</code>	→ Matches the prefix <i>un</i>	undo, unclear, ...
<code>/\Bed\b /</code>	→ Matches the suffix <i>ed</i>	Bed, agreed, freed, greed,
<code>/cat dog /</code>	→ Either-or	cat, dog
<code>/bank(s ing) /</code>	→ Treats <i>ing</i> as single unit	banks, banking
<code>/bank[s ing] /</code>	→ <i>i, n, g</i> as three separate units	banks, banki, bankn, bankg

An example

- Document:

The recent attempt by the police to retain their current rates of pay has not gathered much favor with the southern factions.

- Write the regular expression that find all the occurrences of word '*the*' (case-insensitive).

- RegEx: `/[Tt]he/`

True positives

- Output: *The recent attempt by the police to retain their current rates of pay has not gathered much favor with the southern factions.*

An example

- Document:

The recent attempt by the police to retain their current rates of pay has not gathered much favor with the southern factions.

- Write the regular expression that find all the occurrences of word 'the' (case-insensitive).

- RegEx: `/[Tt]he/`

False positives

Add word boundary in RE

- Output: *The recent attempt by the police to retain their current rates of pay has not gathered much favor with the southern factions.*

An example

- Document:

The recent attempt by the police to retain their current rates of pay has not gathered much favor with the southern factions.

- Write the regular expression that find all the occurrences of word 'the' (case-insensitive).

- RegEx: `/\b[Tt]he\b/`

- Output: *The recent attempt by the police to retain their current rates of pay has not gathered much favor with the southern factions.*

Success!

Regular Expression is more powerful than it seems!

ELIZA [Weizenbaum, 1964-66, MIT AI lab]

- The first chatbot.
- Pattern matching system

User: I am unhappy.

ELIZA: DO YOU THINK COMING HERE WILL HELP YOU NOT TO BE UNHAPPY

User: I need some help, that much seems certain.

ELIZA: WHAT WOULD IT MEAN TO YOU IF YOU GOT SOME HELP

User: Perhaps I could learn to get along with my mother.

ELIZA: TELL ME MORE ABOUT YOUR FAMILY

User: My mother takes care of me.

ELIZA: WHO ELSE IN YOUR FAMILY TAKES CARE OF YOU

User: My father.

ELIZA: YOUR FATHER

User: You are like my father in some ways.

Weizenbaum (1966)

ELIZA

- Substitution with regex
 - s / RE / SubText /
- Three steps:
 - Changing the first person mentions to the uppercase second person mentions
 - I'm → YOU ARE
 - Create a set of all possible substitutions.
 - Rank the possible outputs to respond to the user.

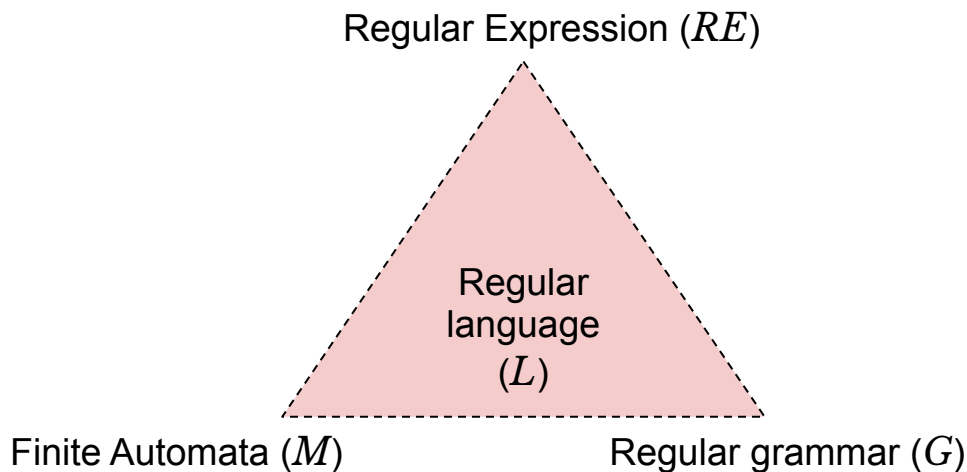
User₁: Men are all alike.
ELIZA₁: IN WHAT WAY
User₂: They're always bugging us about something or other.
ELIZA₂: CAN YOU THINK OF A SPECIFIC EXAMPLE
User₃: Well, my boyfriend made me come here.
ELIZA₃: YOUR BOYFRIEND MADE YOU COME HERE
User₄: He says I'm depressed much of the time.
ELIZA₄: I AM SORRY TO HEAR YOU ARE DEPRESSED

```
s/. * YOU ARE (depressed|sad) .*/ I AM SORRY TO HEAR YOU ARE \1/  
s/. * YOU ARE (depressed|sad) .*/ WHY DO YOU THINK YOU ARE \1/  
s/. * all .*/ IN WHAT WAY/  
s/. * always .*/ CAN YOU THINK OF A SPECIFIC EXAMPLE/
```

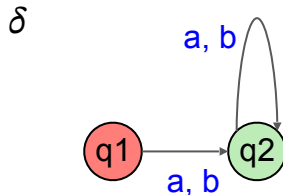
Regular expressions and Finite automata

Regular Expression and Finite Automata

- Regular expressions can be implemented as finite-state automaton.
- Set of strings accepted by regular expression or the corresponding finite automata is termed as regular language.



- $L: \{a, b, ab, aa, ba, aba, \dots\}$
- $RE: (a|b)^+$
- $G:$
 $S \rightarrow a \mid b \mid aS \mid bS$
- $M:$



Finite Automata

- Given an input x , it outputs whether or not the input x is accepted by the automata M .
 - $M(x) \rightarrow \{\text{Accepted, Not accepted}\}$
- Finite Automation
 - Deterministic Finite Automata (DFA)
 - Non-deterministic Finite Automata (NFA)

Deterministic Finite Automata (DFA)

- A finite automata can be represented as quintuple $M = \langle Q, \Sigma, S, F, \delta \rangle$, where
 - Q = Finite set of states
 - Σ = Finite set of input symbols
 - S = Start state $S \subseteq Q$
 - F = Finite set final/accepting states $F \subseteq Q$
 - δ = Transition function. $\delta: q_i \rightarrow_a q_j$, where $q_i, q_j \in Q$ and $a \in \Sigma$

Regular expression: `/[Bb]anks?/`

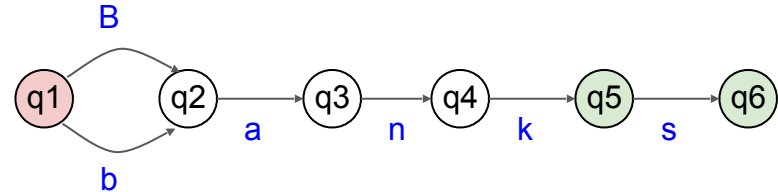
$\Sigma = \{a-zA-Z\}$

$Q = \{q1, q2, q3, q4, q5, q6\}$

$S = \{q1\}$

$F = \{q5, q6\}$

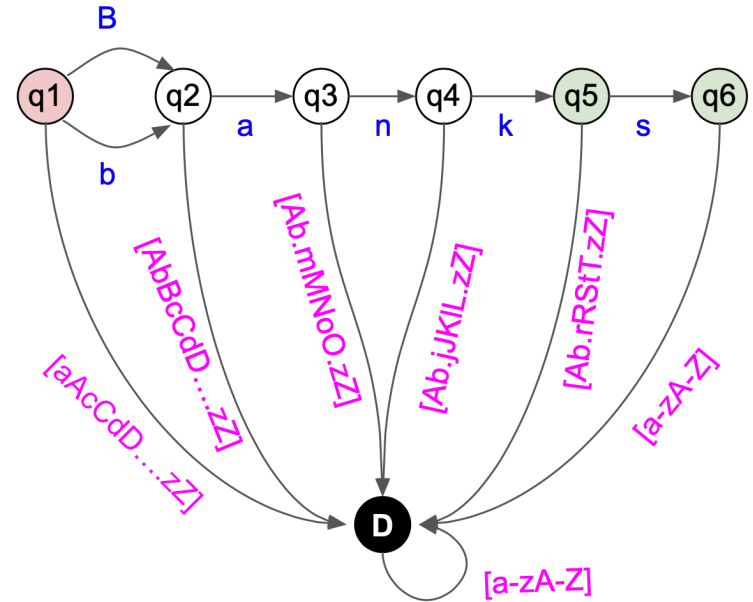
δ :



Transition diagram and table

Q/Σ	a	b	..	k	n	s	..	A	B	...
q1		q2							q2	
q2	q3									
q3					q4					
q4				q5						
q5						q6				
q6										
D										

*All blank entries will point to the dead (D) state



DFA: Example 2

Regular expression: `/baa+!/`

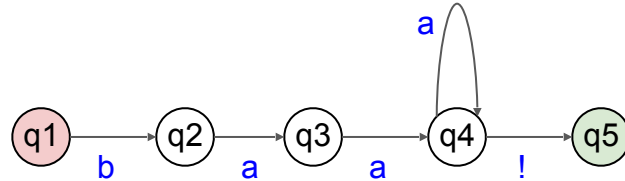
Valid inputs:

baa!

baaa!

baaaaaa!

...



Acceptance procedure:

1. While all the input symbols are not consumed:
 - a. At the current state, if the current input symbol matches any of its outlink labels,
 - i. Consume the input, move along the link to another state, and goto step 1.
 - b. Else
 - i. Announce **failure**
2. If the current state is one of the final states
 - a. Announce **success**
3. Else
 - a. Announce **failure**

Text Processing

Text Processing

- Three most frequent preprocessing steps
 - Sentence segmentation
 - Tokenization
 - Normalization

India has recorded the biggest single-day spike of 66,999 COVID-19 cases, taking the total number of infections to 23,96,637. However, the recovery rate has gone up to 70.76 per cent, with 16,95,982 people recovering in the country from the highly-contagious disease, government data this morning showed. With the death of 942 patients in the last 24 hours, the county's fatality count rose 47,033, the Union Health Ministry said. India is the third worst-hit country by the pandemic after the United States and Brazil.

Sentence Segmentation

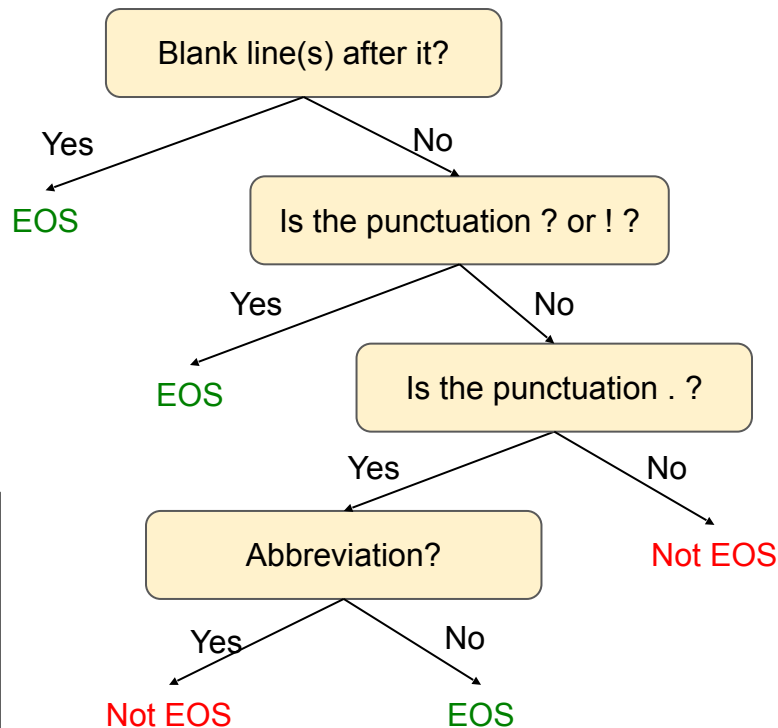
- Usual sentence end-marker: '.', '?', '!'
- Exclamation and Question marks are usually unambiguous.
 - Interjections: Oh!
- Period is quite ambiguous
 - Sentence end marker
 - Acronyms or Abbreviations: Ph.D. , B.Tech , Mr. , Dr.
 - Numbers: 23.11
 - Ellipsis: ...
 - Urls and email id: www.google.co.in, abc@gmail.com

Sentence Segmentation

- Disambiguation
 - Rule-based
 - RegEx based
 - ML based

Some other useful features

- Case of word with “.”: Upper, Lower, Cap, Number
- Case of word after “.”: Upper, Lower, Cap, Number
- Numeric features
 - Length of word with “.”
 - Probability(word with “.” occurs at end-of-s)



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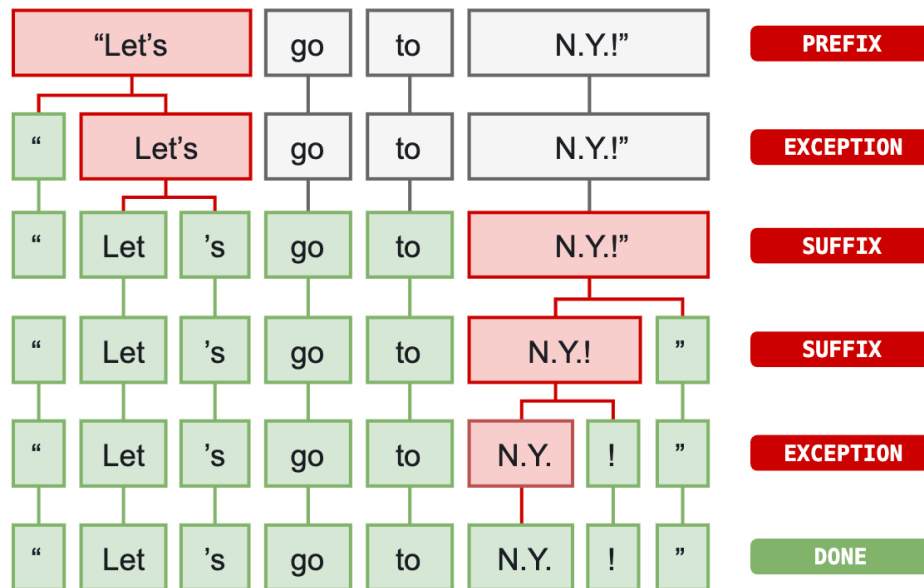
- India has recorded the biggest single-day spike of 66,999 COVID-19 cases, taking the total number of infections to 23,96,637.
- However, the recovery rate has gone up to 70.76 per cent, with 16,95,982 people recovering in the country from the highly-contagious disease, government data this morning showed.
- With the death of 942 patients in the last 24 hours, the county's fatality count rose 47,033, the Union Health Ministry said.
- India is the third worst-hit country by the pandemic after the United States and Brazil.

Tokenization

- Usual delimiters: Whitespace, Dot, Comma, Question mark, Exclamation, etc.
- Some of the special cases
 - Abbreviation: Ph.D., AT&T, Mr., etc.
 - Time and date: 12:30:00, 01.08.2020, or 01/08/2020
 - Apostrophe and clitics: didn't, it's
 - Clitics can not stand on its own.
 - Hyphens: Covid-19, Delhi-based
- Standard for tokenisation
 - LDC Penn Treebank tokenization standard
 - didn't → did + n't
 - it's → it + 's

Tokenization: Spacy

- Rule-based
 - Split the sentences into tokens using whitespace chars (space, tab, etc.)
 - From left to right
 - Does the token requires special attention?
 - Yes
 - Check whether some prefix, suffix, or infix can be split.
 - No
 - continue;



Tokenization: Special cases

- Tokenization in other languages is much more complex:
 - Chinese: No space between words
 - 姚明进入总决赛 (Yao Ming reaches the finals)
 - 姚明(Yao Ming) 进入 (reaches) 总决赛 (the finals) [Chinese Treebank segmentation]
 - Each Chinese character → a single unit of meaning (morpheme) and is pronounceable as a single syllable.

Text Processing

- Three most frequent preprocessing steps
 - Sentence segmentation
 - **Tokenization**
 - Normalization

India has recorded the biggest single-day spike of 66,999 COVID-19 cases, taking the total number of infections to 23,96,637. However, the recovery rate has gone up to 70.76 per cent, with 16,95,982 people recovering in the country from the highly-contagious disease, government data this morning showed. With the death of 942 patients in the last 24 hours, the county's fatality count rose 47,033, the Union Health Ministry said. India is the third worst-hit country by the pandemic after the United States and Brazil.

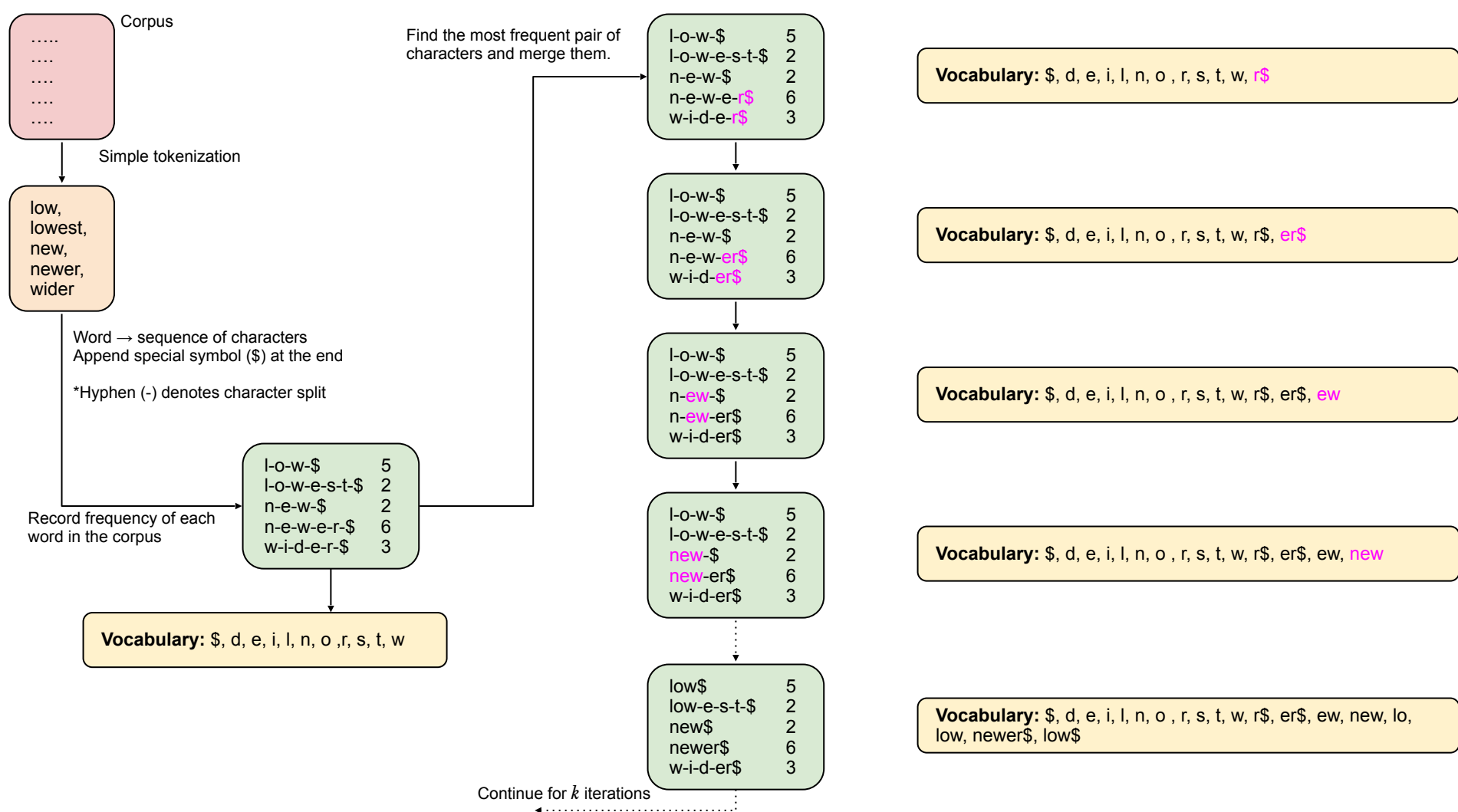
- India has recorded the biggest single - day spike of 66,999 COVID-19 cases , taking the total number of infections to 23,96,637 .
- However , the recovery rate has gone up to 70.76 per cent , with 16,95,982 people recovering in the country from the highly - contagious disease , government data this morning showed .
- With the death of 942 patients in the last 24 hours , the county 's fatality count rose 47,033 , the Union Health Ministry said .
- India is the third worst - hit country by the pandemic after the United States and Brazil .

Unknown or Rare words

- Recall, the purpose of tokenization is to split the sentence into meaningful entities for downstreaming tasks.
- What if there is an unknown word at the inference time? — the system will not have a clue about it.
- Can we do better with the tokenization?
 - Answer: Yes. Subword tokens might help.

Byte-pair Encoding (BPE) <https://arxiv.org/abs/1508.07909>

- Subword-based tokenization
 - Provides a solution to handle unknown or rare words in downstreaming tasks.
- Learn from data what should be tokens.
 - Iteratively merge frequent pairs of characters to form subword/word tokens.



r\$
er\$
ew
new
lo
low
newer\$
low\$

Inference

- Apply merges on the test sentence that we learned in the order
- E.g.,
 - newer
 - n-e-w-e-r-\$ \Rightarrow n-e-w-e-r\$ \Rightarrow n-e-w-er\$ \Rightarrow n-ew-er\$ \Rightarrow new-er\$ \Rightarrow newer\$
 - lower
 - l-o-w-e-r-\$ \Rightarrow l-o-w-e-r\$ \Rightarrow l-o-w-er\$ \Rightarrow lo-w-er\$ \Rightarrow low-er\$

Other Tokenizers

- Word-piece [Schuster et al., 2012] <https://static.googleusercontent.com/media/research.google.com/ja//pubs/archive/37842.pdf>
 - Uses word boundary token at the beginning of the word instead of at the end.
 - Rather than merging two most-frequent pairs, it merges if the pair maximizes the likelihood of the training data once added to the vocabulary.
- Unigram [Kudo, 2018] <https://arxiv.org/pdf/1804.10959.pdf>
 - Initialize the vocabulary to include a large number of symbols — all pre-tokenize words and common substring
 - Iteratively, remove symbols from the vocabulary if the loss is minimal.
- Sentence-piece [Kudo and Richardson, 2018] <https://arxiv.org/pdf/1808.06226.pdf>
 - Instead of pre-tokenize words, it operates on the sequence of characters for a sentence. (Space is a valid symbol)
 - Employs BPE or Unigram tokeniser

Normalization

- Putting words/tokens in a standard format
- Types:
 - Case-Folding
 - *CAT, Cat, cat* ⇒ *cat* or *CAT*
 - Lemmatization: Finding the root form of the word
 - *is, are, was* ⇒ *be*
 - *cat, cats, goose, geese, fly, flies* ⇒ *cat, cat, goose, goose, fly, fly*
 - Stemming: Chopping off the affixes
 - *is, are, was* ⇒ *is, are, wa*
 - *cat, cats, goose, geese, fly, flies* ⇒ *cat, cat, goos, gees, fli, fli*
 - Others
 - Equivalence class
 - *Normalization* vs *Normalisation*, *color* vs *colour*, *center* vs *centre*
 - *U.S.A.* vs *USA* vs *US*

Text Processing

- Three most frequent preprocessing steps
 - Sentence segmentation
 - Tokenization
 - **Normalization: Case-folding**

India has recorded the biggest single-day spike of 66,999 COVID-19 cases, taking the total number of infections to 23,96,637. However, the recovery rate has gone up to 70.76 per cent, with 16,95,982 people recovering in the country from the highly-contagious disease, government data this morning showed. With the death of 942 patients in the last 24 hours, the county's fatality count rose 47,033, the Union Health Ministry said. India is the third worst-hit country by the pandemic after the United States and Brazil.

- india has recorded the biggest single - day spike of 66,999 covid-19 cases , taking the total number of infections to 23,96,637 .
- however , the recovery rate has gone up to 70.76 per cent , with 16,95,982 people recovering in the country from the highly - contagious disease , government data this morning showed .
- with the death of 942 patients in the last 24 hours , the county 's fatality count rose 47,033 , the union health ministry said .
- india is the third worst - hit country by the pandemic after the united states and brazil .

Text Processing

- Three most frequent preprocessing steps
 - Sentence segmentation
 - Tokenization
 - **Normalization: Lemmatization**

India has recorded the biggest single-day spike of 66,999 COVID-19 cases, taking the total number of infections to 23,96,637. However, the recovery rate has gone up to 70.76 per cent, with 16,95,982 people recovering in the country from the highly-contagious disease, government data this morning showed. With the death of 942 patients in the last 24 hours, the county's fatality count rose 47,033, the Union Health Ministry said. India is the third worst-hit country by the pandemic after the United States and Brazil.

- india **have record** the **big** single - day spike of 66,999 covid-19 **case** , **take** the total number of **infection** to 23,96,637 .
- however , the recovery rate **have go** up to 70.76 per cent , with 16,95,982 people **recover** in the country from the highly - contagious disease , government data this morning **show** .
- with the death of 942 **patient** in the last 24 hour , the county 's fatality count **rise** 47,033 , the union health ministry **say** .
- india **be** the third worst - hit country by the pandemic after the united state and brazil .

Text Processing

- Three most frequent preprocessing steps
 - Sentence segmentation
 - Tokenization
 - **Normalization: Stemming**

India has recorded the biggest single-day spike of 66,999 COVID-19 cases, taking the total number of infections to 23,96,637. However, the recovery rate has gone up to 70.76 per cent, with 16,95,982 people recovering in the country from the highly-contagious disease, government data this morning showed. With the death of 942 patients in the last 24 hours, the county's fatality count rose 47,033, the Union Health Ministry said. India is the third worst-hit country by the pandemic after the United States and Brazil.

- india **ha record** the biggest single - day spike of 66,999 covid-19 **case** , **take** the total number of **infect** to 23,96,637 .
- **howev** , the **recoveri** rate **ha** gone up to 70.76 per cent , with 16,95,982 **peopl recov** in the **countri** from the highly - **contagi** **diseas** , **govern** data **thi morn show** .
- with the death of 942 **patient** in the last 24 **hour** , the **counti** 's **fatal** count rose 47,033 , the union health **ministri** said .
- india is the third worst - hit **countri** by the **pandem** after the **unit state** and brazil .

Thanks